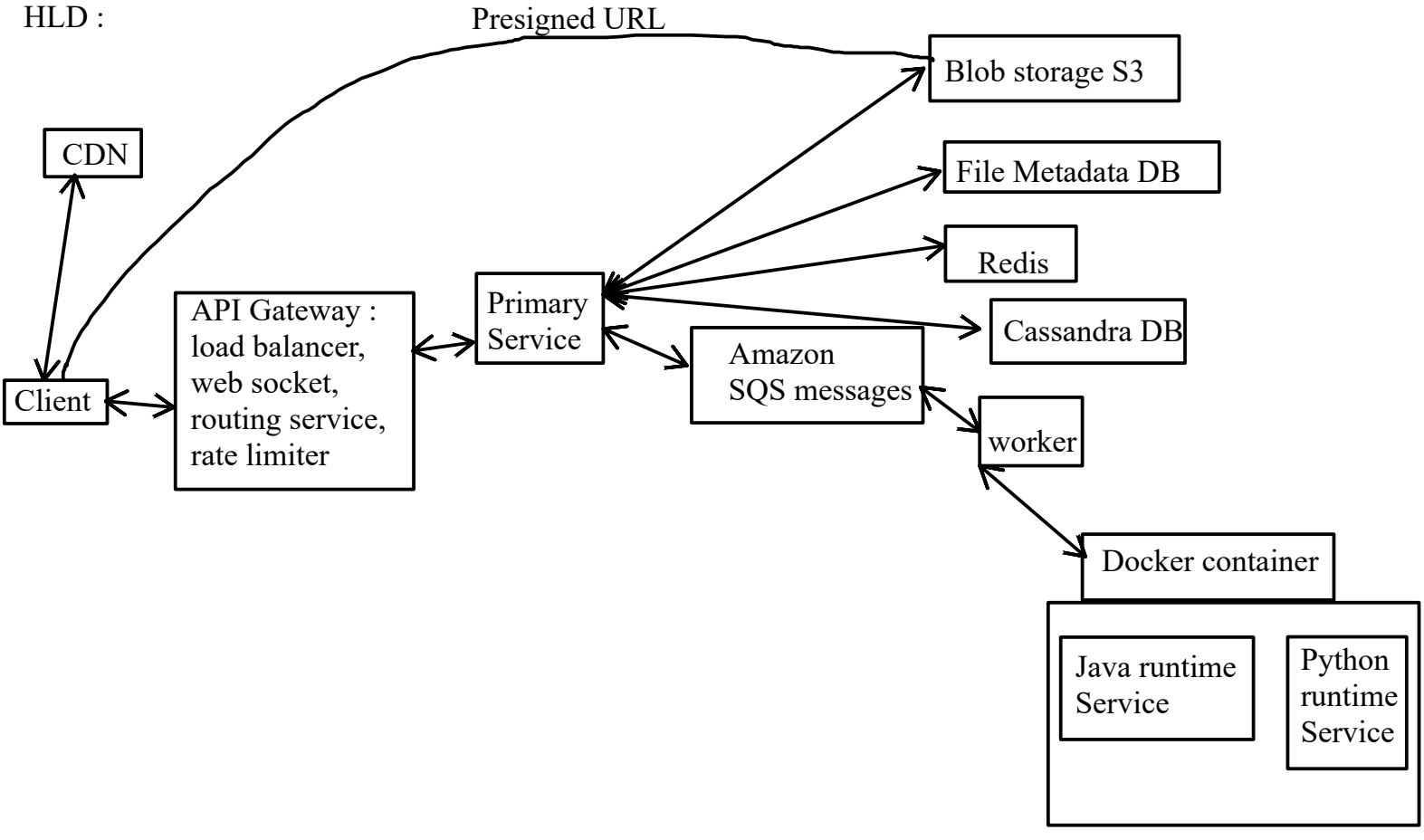
Leetcode

Functional requirements:
user should be able to search for any problem & submit solution based on his preference,
user should be able to participate in any coding contest based on his preference,
rank of user should get updated after every 1second

Non-functional requirements:
Data-partitioning,
load balancig,.
fault tolerance,
vertical scalabaility,
horizontal scalability,
low latency,
high throughput,
CAP theorem(Here we need Strong consistency for rank predictions during contest & for search scenario eventual consistency is fine i.e 99.99% uptime i.e Availability>>Consistency)

Core Entities:
user,
problem,
contest,
solution

API :
POST:solution/submit
{
userID,
{filemetadata}
}
GET:/search/{problemID}

HLD :